THE UNIVERSITY OF SYDNEY

July 1, 2022

Dear Editors of *Nature Computational Science*,

We are pleased to submit our article, *"Unifying Pairwise Interactions in Complex Dynamics"*, for consideration for publication in *Nature Computational Science*.

The problem of quantifying interactions in complex dynamical systems is a fundamental scientific problem that has stimulated decades of theory and has enabled important breakthroughs in quantifying and understanding patterns in brain dynamics, climate systems, financial markets, and many other systems. Different scientific literatures—from economics (Granger causality), information theory (mutual information), audio analysis (dynamic time warping), to climatology (convergent cross mapping)—define interactions between time series differently, resulting in a large and disjoint methodological literature. Importantly, *these methods for quantifying and understanding complex dynamical systems remain unstructured: they have never previously been unified and compared*. This means that a scientist aiming to analyze their time-series dataset has no principled way to draw on this rich, interdisciplinary literature of scientific methods.

In this paper we unify 249 diverse scientific methods for quantifying pairwise interactions from time series for the first time, an exercise that is unprecedented in its scale and interdisciplinary breadth. We then take a data-driven approach to organizing these methods, by first assembling a multidisciplinary library of over 1000 diverse (simulated and real-world) multivariate time series, and then analyzing how similarly all methods behave on these data. The result is an incredibly powerful data-driven organization of diverse scientific methods, that: (i) unifies seemingly disparate scientific methods into interpretable high-level conceptual groupings; (ii) automatically recapitulates recently published connections between different methods; and (iii) suggests new and important relationships, bridging otherwise disconnected scientific domains and their findings.

We go on to show how scientists can draw on the combined literature to solve real problems involving complex time series in new and powerful ways. We present a case study of movement classification from smart-watch data and demonstrate that, instead of manually selecting a single scientific analysis method, the most accurate method from across the scientific literature can be discovered automatically, yielding interpretable insights into the dataset's structure. Furthermore, by drawing on the full interdisciplinary literature of 249 methods, we report even higher performance than any individual method, demonstrating the power of solving problems by simultaneously drawing from multiple scientific methods.

The new connections we make between diverse theoretical literatures are likely to have wide-ranging impact on the development and use of methods to understand complex systems. In addition, our article is accompanied by two impactful resources for the study of complex systems: (i) a data library of $>$1000 diverse multivariate time series; and (ii) a flexible and extendable open-source python package, `pyspi`, which implements our full library of 249 scientific methods. Broad interest in this work is clear from the strong response to our arXiv pre-print, which has been widely shared (e.g., $>$ 270 retweets) by neuroscientists, ecologists, sociologists, and complex-systems scientists (among others), and becoming the top research article on social media the week that it was released (according to paperswithcode).

Thank you for your consideration of our article,

Oliver Cliff, Joseph Lizier, Naotsugu Tsuchiya, Ben Fulcher.